

Solved selected problems of Classical Electrodynamics - Hans Ohanian

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Chapter 6 - The Theory of Special Relativity

Exercises

Solution. Exercise 1.

Let us consider the round-trip of the parallel beam from b to d_1 to b . If L is the distance from b to d_1 then the time taken to go from b to d_1 is

$$t_1 = \frac{L}{c - V}$$

Since the velocity of the beam in that case is $c - V$. In the same way, for the trip back from d_1 to b the time taken is

$$t_2 = \frac{L}{c + V}$$

The average speed is given by $\bar{v} = 2L/(t_1 + t_2)$ so

$$\begin{aligned}\bar{v} &= \frac{2L}{t_1 + t_2} = \frac{2L}{\frac{L}{c-V} + \frac{L}{c+V}} = \frac{2}{\frac{c-V+c+V}{(c-V)(c+V)}} = \\ &= \frac{(c-V)(c+V)}{c} = \frac{c^2 - V^2}{c} = c \left(1 - \frac{V^2}{c^2}\right)\end{aligned}$$

If the average speed for the perpendicular beam is given by $\sqrt{c^2 - V^2} = c\sqrt{1 - V^2/c^2}$ then the time a round-trip takes is

$$t_{\perp} = \frac{2L}{c\sqrt{1 - V^2/c^2}}$$

So the time delay Δt between the parallel and the perpendicular beam is

$$\begin{aligned}\Delta t &= t_{\parallel} - t_{\perp} \\ &= \frac{2L}{c(1 - V^2/c^2)} - \frac{2L}{c\sqrt{1 - V^2/c^2}} \\ &= 2 \left[\frac{L}{c(1 - V^2/c^2)} - \frac{L}{c\sqrt{1 - V^2/c^2}} \right]\end{aligned}$$

Also, we know the frequency of the beams is $1/T = c/\lambda$ then the phase difference must be

$$\Delta\phi = \frac{\Delta t}{T} = 2 \left[\frac{L}{c(1 - V^2/c^2)} - \frac{L}{c\sqrt{1 - V^2/c^2}} \right] \frac{c}{\lambda}$$

Finally, assuming $V^2/c^2 \ll 1$ we may approximate the denominators by

$$\begin{aligned} \Delta\phi &\approx 2 \left[\frac{L}{c}(1 + V^2/c^2) - \frac{L}{c}(1 + V^2/2c^2) \right] \frac{c}{\lambda} \\ &\approx 2L \left[\frac{V^2}{c^2} - \frac{V^2}{2c^2} \right] \frac{1}{\lambda} \\ &\approx \frac{L}{\lambda} \frac{V^2}{c^2} \end{aligned}$$

□

Solution. Exercise 2.

The time between crests being emitted is $\tau_s = 1/\nu_s$ and hence the source will move during that time a distance $(\hat{\mathbf{n}} \cdot \mathbf{V}_s)\tau_s$.

If we assume the light is being emitted to the right while the source is moving to the right then the previous crest emitted will have moved a distance λ from the source reference frame.

Therefore the distance between crests emitted to the right from the ether reference frame is

$$\lambda' = \lambda - (\hat{\mathbf{n}} \cdot \mathbf{V}_s)\tau_s$$

So the frequency of the light from the ether reference frame is

$$\begin{aligned}\nu &= \frac{c}{\lambda'} \\ &= \frac{c}{\lambda - (\hat{\mathbf{n}} \cdot \mathbf{V}_s)\tau_s} \\ &= \nu_s \frac{c}{\lambda\nu_s - \hat{\mathbf{n}} \cdot \mathbf{V}_s} \\ &= \nu_s \frac{1}{1 - \hat{\mathbf{n}} \cdot \mathbf{V}_s/c}\end{aligned}$$

Where we used that $\lambda\nu_s = c$ and hence

$$\nu_s = \nu \left(1 - \frac{\hat{\mathbf{n}} \cdot \mathbf{V}_s}{c} \right)$$

Let us suppose now that the receiver is "chasing" the source with a velocity $\mathbf{V}_r$, then the time between crests arriving to the receiver is $\tau_r = 1/\nu_r$, during that time the receiver will have moved a distance $(\hat{\mathbf{n}} \cdot \mathbf{V}_r)\tau_r$.

Therefore the distance between received crests from the ether reference frame is

$$\lambda' = \lambda - (\hat{\mathbf{n}} \cdot \mathbf{V}_r)\tau_r$$

So the frequency of the light from the ether reference frame is

$$\begin{aligned}\nu &= \frac{c}{\lambda'} \\ &= \frac{c}{\lambda - (\hat{\mathbf{n}} \cdot \mathbf{V}_r)\tau_r} \\ &= \nu_r \frac{c}{\lambda\nu_r - \hat{\mathbf{n}} \cdot \mathbf{V}_r} \\ &= \nu_r \frac{1}{1 - \hat{\mathbf{n}} \cdot \mathbf{V}_r/c}\end{aligned}$$

Where we used that also $\lambda\nu_r = c$ and hence

$$\nu_r = \nu \left(1 - \frac{\hat{\mathbf{n}} \cdot \mathbf{V}_r}{c} \right)$$

Finally, joining both equations for ν_s and ν_r we get that

$$\nu_r = \frac{\nu_s}{1 - \frac{\hat{\mathbf{n}} \cdot \mathbf{V}_s}{c}} \left(1 - \frac{\hat{\mathbf{n}} \cdot \mathbf{V}_r}{c} \right)$$

$$\nu_r = \nu_s \frac{1 - \hat{\mathbf{n}} \cdot \mathbf{V}_r / c}{1 - \hat{\mathbf{n}} \cdot \mathbf{V}_s / c}$$

□

Solution. Exercise 3.

We know that $dy' = dy$ so

$$u'_y = \frac{dy'}{dt'} = \frac{dy}{\gamma(dt - Vdx)} = \frac{(dy/dt)\sqrt{1 - V^2}}{1 - V(dx/dt)} = u_y \frac{\sqrt{1 - V^2}}{1 - u_x V}$$

In the same way, since $dz' = dz$ we get that

$$u'_z = \frac{dz'}{dt'} = \frac{dz}{\gamma(dt - Vdx)} = \frac{(dz/dt)\sqrt{1 - V^2}}{1 - V(dx/dt)} = u_z \frac{\sqrt{1 - V^2}}{1 - u_x V}$$

□